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Development and validation of Visual Impairment as a Risk for Falls Questionnaire

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Abstract

Purpose: visual impairment is associated with an increased risk of falls, yet eye care professionals are infrequently members of falls prevention clinics. The aim of this preliminary study was to validate a newly created Visual Impairment as a Risk for Falls Questionnaire intended to be used by those professionals not involved in eye care.
Methods: about 53 participants with various visual impairments known to be associated with falls and 33 participants with normal sight were contacted within 4 months of a full oculo-visual assessment and were asked the questions from the current questionnaire pertaining to their visual function. A retrospective file review was undertaken and the sensitivity and specificity of participants' responses were calculated compared to the actual vision impairment based on the findings from their visual assessment.
Results: the question regarding ability to read was included to identify people with central vision loss, a risk factor for falling. It was found to have sensitivity of 74% and specificity of 87%. Both sensitivity and specificity improved when participants with cognitive impairment were excluded. The question on recognising facial features gave sensitivity of 73% and specificity of 97% for this subgroup. However, questions related to impairments in stereopsis and peripheral fields were not useful.
Conclusion: the study demonstrates that several questions of the new questionnaire are useful; however, further testing with a larger population is needed to fully validate the questionnaire for use by health care professionals.

Keywords: *low vision, falls, fall prevention questionnaire, screening, older people*

Background

It has been well established that advancing age is a risk factor for falling. In fact, 28–35% of adults over 65 years, experience a fall each year [1]. The impact of a fall can be disastrous, leading to both physical and psychological impairment and increased mortality [2–5]. With an aging population, the number of older individuals falling is likely to increase, necessitating the need to prevent the occurrence of falls. Age, however, is not the sole risk factor for a fall, with the underlying cause often being multifactorial in nature [1, 4, 6–8]. Vision plays an integral role in maintaining posture and balance. This is evidenced in the 'subway platform' phenomenon; wherein an individual's peripheral view of a moving object leads to the illusion of self-motion, or through increasing postural sway by removing vision simply by closing one's eyes [9]. A deficiency in vision can lead to an increased risk of a fall [1, 2, 6, 7]. Impairment in visual acuity, stereopsis, contrast sensitivity or visual field has been consistently reported to increase the likelihood of falling [10]. Visual acuity refers to an individual's ability to distinguish objects at a specific distance [11]. Contrast sensitivity refers to an individual's ability to distinguish lower contrast objects. Stereopsis is the ability to see objects in depth and visual field is a measure of the extent of peripheral vision. An unsuitable spectacle prescription has also been identified as a risk factor for falling [3, 10]. In order to address the multifactorial nature of a fall, multidisciplinary fall prevention clinics have been created. Eye care professionals, however, are often not consulted in Canada and possibly in other countries, and are seldom invited as participating members of the multidisciplinary clinic team.

The American Geriatrics Society/British Geriatrics Society (AGS/BGS) Clinical Practice Guideline for the prevention of falls in older persons dictates that 'Patients should be asked to identify and describe any vision-related problems and concerns. If they report problems or concerns, their vision should be formally assessed and any remediable visual abnormalities should be treated, particularly cataracts [12].' While several questionnaires detailing visual function currently exist, they are

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typically lengthy [14, 16] and unlikely to be adopted into another health care professional's compendium of necessary intake questions involved in a systems review. Other questionnaires have one or two questions that address only one visual function measure [13, 15]. The Visual Impairment as a Risk for Falls Questionnaire (VIRFQ) was developed as a quick tool to assist in the recognition of a visual risk for a fall. It is designed to be used by non-vision-related health care providers in a mobility and fall prevention clinic and for other practitioners who encounter a patient with other risk factors for falling, to assist with triage and appropriate referral for eye care. The purpose of this preliminary study was to develop/compile and validate the questionnaire, ensuring that the questions chosen accurately identified a visual impairment.

Method

This cross-sectional study received clearance through a University of Waterloo Research Ethics Committee and adheres to the principles of the Declaration of Helsinki. The questionnaire was developed from existing questionnaires with the addition of one newly created question. Three initial background questions (A–C, see Figure 1) were selected to determine the date of the last eye exam, and spectacle use. Questions A and B were adapted from the Saskatoon Health Region Fall—Risk Screening: Multi-Factorial Questionnaire [13], and the Desmond Fall Risk Questionnaire [14], respectively, and C was newly created. Questions 2–7 were chosen and revised in order to attempt to accurately identify impaired visual function measures that are deemed risk factors for falling. Questions 2–6 were modified from similar questions found in the National Eye Institute Visual Functioning Questionnaire-25 (VFQ-25) [14]. The third question was adapted from both the Saskatoon Questionnaire and the VFQ-25 [13, 14]. Question one was an altered version of that found in the Desmond Fall Risk Questionnaire [15]. The last question was derived from the Independent Mobility Questionnaire [16]. The questions were prefaced by the prompt 'with your glasses on' to serve as a reminder that the response should be based on their habitual vision rather than the need for glasses. It does not preclude those who do not wear glasses from responding. Aside from the preliminary questions A–C, all questions required a yes/no response. A copy of the questionnaire is shown in Figure 1.

Participants with and without a visual impairment were recruited consecutively from the Waterloo Optometry and Vision Science (WOVS) Primary Care (general care), Ocular Health (ocular disease management) and Low Vision (vision rehabilitation) Clinics by a research assistant. To ensure that it was possible to validate questions identified those with particular visual function impairments, the sample was deliberately enriched with those with visual impairment by recruiting from the Ocular Health and Low Vision Clinic. The study participants were adults over the age of 65 years and who were able to speak English. Both female and male participants were invited. The participants had had a full eye exam at WOVS within the previous four months. Individuals who agreed to participate were provided information about the study and if consent obtained, they were interviewed by phone and asked to answer the questions of the VIRFQ and the TICS-M (Telephone Interview Cognitive Status-Modified) [17]. The telephone interviews were conducted between May and August 2013 and 2014. A retrospective chart review was then conducted. The individual's age, last complete eye examination, type of spectacle use (e.g. single vision, bifocals, progressive lenses), best-corrected distance and near visual acuities, contrast sensitivity, stereoacuity, and visual field assessments (central and/or peripheral) were retrieved. Participants who had some form of visual impairment were grouped into three impairment categories: Group 1. Visual acuity (VA) < 0.1 logMAR (<20/25) in the better eye, OR contrast sensitivity (CS) < 1.4, OR central field defect, Group 2. Asymmetric acuities >0.1 logMAR difference between the eyes or an ocular deviation and Group 3. Peripheral field defect. Participants were considered to have a central defect if they reported metamorphopsia, blur or scotomas on Amsler testing and were considered to have a peripheral field impairment if documented as 'constriction' on confrontation visual fields, or having a defect of any type (constriction/step/hemianopia/quadrantanopia) identified via perimetry.

Preliminary questions

A. When was your last eye exam?
B. Do you wear bifocals, trifocals or progressive bifocals (no-line bifocals)?
C. Do you walk while wearing them?

For the following questions, please answer for when you are wearing your regular glasses.

1. With your glasses on, is your vision notably better in one eye compared to the other?
2. With your regular reading glasses or bifocals on, do you have difficulty reading the newspaper?
3. Because of your eyesight, do you have difficulty noticing objects off to the side?
4. Because of your eyesight, do you have difficulty going down steps or stairs or have trouble seeing a curb?
5. With your glasses on, are you able to distinguish a person's facial features from across a room?
6. With your glasses on, do you have difficulty seeing the television?
7. With your glasses on, do you trip over objects because you do not see them?

Figure 1. Visual impairment as a risk for falls questionnaire.

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Table 1. Sensitivity and specificity results of Visual Impairment as a Risk for Falls Questionnaire

	Q1 Sens #/N % (95% CI) Spec #/N % (95% CI)	Q2 Sens #/N % (95% CI) Spec #/N % (95% CI)	Q3 Sens #/N % (95% CI) Spec #/N % (95% CI)	Q4 Sens #/N % (95% CI) Spec #/N % (95% CI)	Q5 Sens #/N % (95% CI) Spec #/N % (95% CI)	Q6 Sens #/N % (95% CI) Spec #/N % (95% CI)	Q7 Sens #/N % (95% CI) Spec #/N % (95% CI)	All Sens #/N % (95% CI) Spec #/N % (95% CI)
Group 1 (central loss)		23/31 74% (55–86) 50/58 86% (74–93)		13/31 46% (25–61) 52/58 90% (78–96)	16/31 52% (33–69) 57/58 98% (90–99.9)	15/31 48% (31–67) 56/58 97% (87–94)		
Group 1 TICs-M >21 (central loss)		13/15 87% (58–98) 34/37 92% (77–98)		6/15 40% (18–67) 32/37 87% (70–95)	11/15 73% (45–91) 36/37 97% (84–99.9)	10/15 67% (39–87) 36/37 97% (84–99.9)		
Group 2 (asymmetric acuities)	13/44 57% (35–76) 16/45 56% (36–73)							
Group 2 TICs-M >21 (asymmetric acuities)	13/23 57% (35–76) 16/29 56% (36–73)							
Group 3 (peripheral loss)			5/10 50% (20–80) 65/79 86% (83–72)	4/10 40% (14–73) 64/79 81% (70–89)			2/10 20% (3.5–56) 68/79 86% (76–93)	
Group 3 TICs-M >21 (peripheral loss)			38% 3/8 38% (10–74)	3/8 38% (10–74)			1/8 13% (0.07–53)	
			82% 36/44 82% (67–91)	36/44 82% (67–91)			36/44 82% (67–91)	
Aggregate Impairment								53/60 88% (77–95) 14/29 48% (30–67)
Aggregate Impairment TICS-M >21								26/52 87% (68–96) 12/22 55% (32–75)

Sens, sensitivity; Spec, specificity; TICS-M, Telephone Interview Cognitive Status-Modified; CI, confidence interval.
Group 1: VA > 0.1 logMAR, OR CS < 1.40 OR central field defect.
Group 2: VA difference >0.1 logMAR.
Group 3: Peripheral field loss.
Aggregate Impairment: any impairment responding yes to any question.

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Analysis: The sensitivities and specificities were calculated treating the above three impairment categories as the gold standard diagnosis and the VIRFQ questions (1–7) as the test predictors. See Table 1 for grouping of questions relevant for each category. For question 5, an impairment was indicated if a 'no' response was given because of the direction of the question, instead of the 'yes' response to all the other questions. The 2 × 2 tables were constructed for each impairment category versus questions selected as detectors of the impairment, and the sensitivity and specificity (with 95% confidence intervals) of each question were calculated. Questions 2 and 4–6 were analysed for Group 1 participants, Question 1 was analysed for Group 2 and Questions 3, 4 and 7 were evaluated for Group 3. It is possible that an individual with asymmetric acuities would also fall into the group with reduced acuity or peripheral vision loss. They were therefore analysed in both groups. Participants were further subdivided to identify responses of participants with a TICS-M score >21 and the same analysis undertaken for each group separately to determine if the questionnaire was more valid for those without a cognitive impairment. Finally, all the questions were evaluated as an aggregate (a response to any one question being a positive response) for detecting any visual impairment as defined above (including Groups 1, 2 and 3).

Results

Ninety-three individuals participated in the study. One participant was excluded because s/he did not wish to answer the TICS-M. Two were excluded as they did not complete the questions of the VIRFQ and one declined to answer any questions after agreeing to participate. The final sample included 89 participants, aged 65–96. Age and TICS-M scores are described for visually impaired and not visually impaired individuals according to each impairment group in Table 2. About 65 (73%) of participants wore some form of multifocal spectacle prescription with 60 (67%) stating that they wore the glasses while walking.

Question number 2 yielded the most promising sensitivity (74%) and specificity (86%) values for Group 1 impairments. These values increased when considering only those with a TICS-M > 21–87% and 92%, respectively. Question 5 also gave good values with a sensitivity of 73% and specificity of 97% for detecting Group 1 impairment. The remaining questions had a high specificity (90–98%) but low sensitivity. Question 1 for Group 2 against question 1, gave moderate sensitivity and specificity (68/58%) which decreased when considering those only with TICS-M > 21. Questions 3, 4 and 7 for Group 3 gave high specificity (81–86%) but lower sensitivity. This did not change significantly for those with TICS-M > 21.

Questions 1–7 in aggregate (a positive response from any question) for detecting any visual impairment, gave high sensitivity (88%) but the specificity was low (48%) and did not change significantly when including only those with TICs-M > 21.

Discussion

The VIRFQ was developed to assist health professionals not involved in eye care to identify individuals who may be at risk of falling due to suboptimal vision or visual appliances. The initial preliminary (A) question was included as the best intervention is prevention. The patient should be referred for an eye examination if one was not conducted within the past year. The prevalence of ocular disease increases with age but many ocular diseases do not present with symptoms at the onset, necessitating regular routine eye examinations [4]. In the older age group, yearly routine eye examinations are effective at detecting new eye diseases and prescription changes [18]. Prompt extraction of first eye cataract has been demonstrated to decrease the falls risk [19]. Inappropriate spectacle prescription has also been linked with an increased falls risk [10]. Older individuals should have the most up to date spectacle prescription, taking care that the prescription is not changed greatly to allow for adaptation [3]. For active individuals, there should be consideration of separate distance glasses for walking [20] and other similar activities as the bifocal add portion blurs the area where a person steps and also alters depth perception [21]. Attention should be given to lens tint to reduce

Table 2. Patient demographics by impairment grouping

	N (VI)	*N* (NV)	Age (VI)	Age (NV)	Avg TICS-M (VI)	Avg TICS-M (NV)
Group 1 (central loss)	31	58	83.61 ± 7.26	74.4 ± 6.28	19.8 ± 5.45	23.74 ± 5.87
Group 1 TICs-M >21 (central loss)	15	37	82.27 ± 7.43	73.0 ± 6.04	24.67 ± 2.09	27.13 ± 4.03
Group 2 (asymmetric acuities)	44	45	79.02 ± 8.14	76.22 ± 7.52	21.16 ± 5.78	23.56 ± 6.02
Group 2 TICs-M >21 (asymmetric acuities)	23	29	77.57 ± 7.52	74.17 ± 7.54	25.57 ± 3.44	27.1 ± 3.84
Group 3 (peripheral loss)	10	79	75.6 ± 6.45	77.86 ± 8.09	23.3 ± 2.86	22.19 ± 6.29
Group 3 TICs-M >21 (peripheral loss)	8	44	77.5 ± 5.63	75.34 ± 7.99	24.62 ± 2.6	26.75 ± 3.83
Aggregate Impairment	60	29	79.45 ± 7.97	73.67 ± 6.06	21.2 ± 5.64	24.69 ± 6.09
Aggregate Impairment TICS-M >21	30	22	78.17 ± 7.72	72.09 ± 5.77	25.68 ± 3.62	27.32 ± 3.79

VI, visually impaired; NV, normal vision; VA, visual acuity; TICS-M, Telephone Interview Cognitive Status-Modified; Avg, average.
Group 1: VA > 0.1 logMAR, OR CS < 1.40 OR central field defect.
Group 2: VA difference >0.1 logMAR.
Group 3: Peripheral field loss.

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photosensitivity and glare, and frame size to avoid obstruction of peripheral field. The other two preliminary questions (B,C) were included as individuals over the age of 65 should be referred to an optometrist for further evaluation if they report that they walk with bifocal type lenses, and have not discussed this with their optometrist.

There are several studies that demonstrate that a decrease in visual acuity is a falls risk [4]. The Beaver Dam Study found that those over the age of 60 with a habitual acuity of 20/25 or less were 2.6 times more likely to have multiple falls [22] while the Blue Mountain Study stated that an acuity of less than 20/30 was associated with 1.9× greater risk of multiple falls [23]. Currently there is no established cut-off for acuity loss associated with an increased risk of falling [4]. Contrast sensitivity is also believed to be a significant falls risk as studies show that it is a good predictor of postural stability and mobility [24]. Individuals with a contrast sensitivity of less than 1.4 are considered to have an early impairment affecting the ability to read, and as such, this was chosen as the limit for CS loss in this study [25]. As with acuity, there is no established degree of contrast sensitivity loss associated with increased prevalence of falls [4] and it is difficult to isolate reduced contrast sensitivity from visual acuity loss when there is a central defect. Therefore, impairments in acuity, contrast sensitivity and central defects were grouped together. The results indicate that question 2 effectively identifies individuals with these impairments as well as possibly question 5 for those with TICs-M > 21; however, a larger sample size would provide stronger evidence.

When one considers vision and falling, it would seem obvious that a loss within the inferior visual field would lead to an increase in falls due to the individual's ability to detect trip hazards, and this holds true [4]. However, those who have head injuries as a result of other forms of peripheral field loss [26] or central field impairment [27] are also more likely to fall. Unfortunately, in our study sample, there was an insufficient number of individuals with measured peripheral visual field loss (n = 10) to accurately determine whether the questions chosen were valid. Further research with specific types of peripheral field defects is required.

The final visual measure often identified as being a risk factor for falls is impairment in stereopsis. Lord and Dayhew concluded that stereopsis was the greatest visual risk factor for multiple falls [10]. In that study, those with asymmetric acuities, defined as at least one line of difference between eyes, had a similar risk for falls as those with poor visual acuity in the better eye [10]. Felson *et al.* showed a higher risk for those who had a similar inter-ocular difference [28]. Question 1 was evaluated for this group, however, the specificity and sensitivity using this criterion was not promising.

Interestingly, the two questions (numbers 2 and 5) which were found to have the best sensitivity and specificity were those adapted from the VFQ-25 [13] and used in the 2006 PALs survey of Canadians who identify as having a seeing disability [29].

A few limitations to this study were identified. A greater sample size for those with visual field impairment is recommended. Data utilised in the study was routinely collected clinical data. Thus visual fields and stereopsis were not measured on many patients. Additionally identification of visual field defects relied largely on confrontation testing, leading to a less accurate classification. Also, the assumption was made that participants were using the most up to date prescription as recorded in their clinical file which would yield optimal acuities, but this may not always be the case. Finally, The WOVS clinic is in an area of slightly above average socioeconomic status compared to the province of Ontario but [30] the clinic population is similar in terms of patient age and sex distribution to a nationwide survey of Canadian optometric practices [31].

Conclusion

While this study demonstrated that those with a reduced visual acuity, contrast sensitivity and central defect would be accurately identified by the VIRFQ, further testing with a greater sample size is required using a research population with controlled measures of visual function. The preliminary three questions would be included, to identify those who would benefit from an eye examination and advice regarding their type of spectacles lenses. A useful, yet succinct visual risk for falls questionnaire would prove helpful in identifying individuals who have an increased falls risk due to vision loss.

Key points

- Visual impairment is a risk factor falls.
- Visual impairment should be considered when assessing a patient who is at risk for falling or has a history of falls.
- The Validation of Visual Impairment as a Risk for Falls Questionnaire (VIRFQ) is a questionnaire that helps identify those who have an impairment in acuity, contrast sensitivity and central field.
- Additional research is required to further develop the questionnaire to identify those with impairments in peripheral vision and depth perception.

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Conflict of interest

None declared.

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